

$$\begin{array}{c}
 a^T: \text{row} \\
 1 \times 3 \\
 \boxed{1 \quad 2 \quad 3}
 \end{array}
 \times
 \begin{array}{c}
 b: \text{column} \\
 3 \times 1 \\
 \boxed{4} \\
 \boxed{5} \\
 \boxed{6}
 \end{array}
 =
 \begin{array}{c}
 \text{a scalar} \\
 1 \times 1 \\
 \boxed{32}
 \end{array}$$

$1 \times 4 + 2 \times 5 + 3 \times 6$
 $= 4 + 10 + 18 = 32$

inner sizes must match ($3 = 3$); the outer sizes give the result's shape (1×1)